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GOSIM Shenzhen 2026

构建自进化与长程任务智能体

Building Self-Improving and Long-Horizon Agents

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01

Introduction to Agent Harness

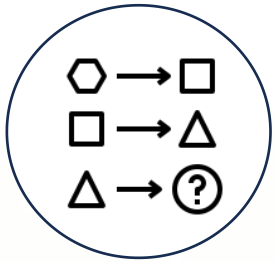


The Five Levels of AGI



Level 1: Chatbots

Conversational AI



Level 2: Reasoners

Human-Level Problem Solving



Level 3: Agents

Autonomous AI

Level 4: Innovators

Innovating AI

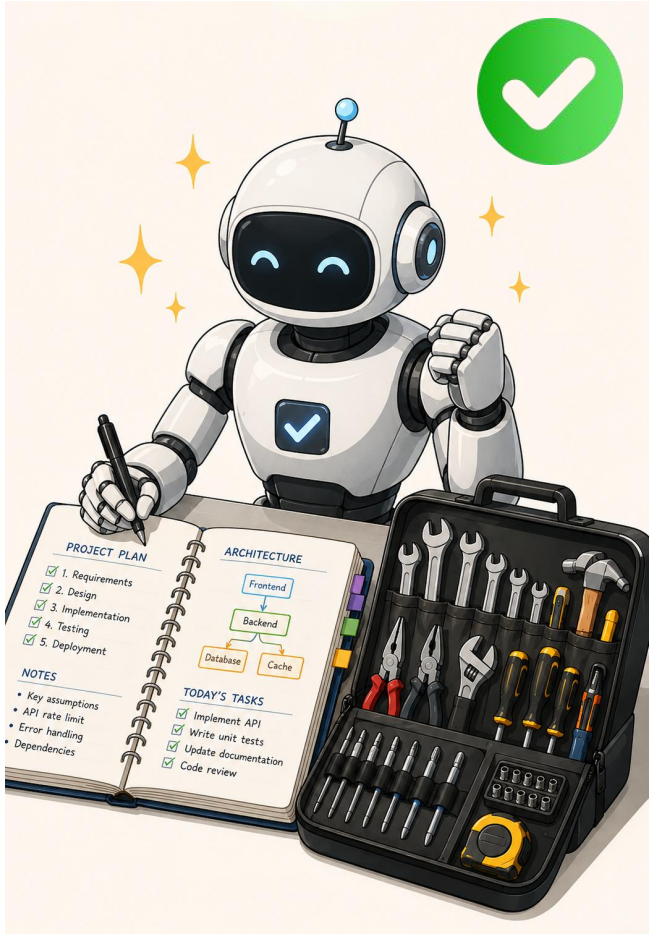


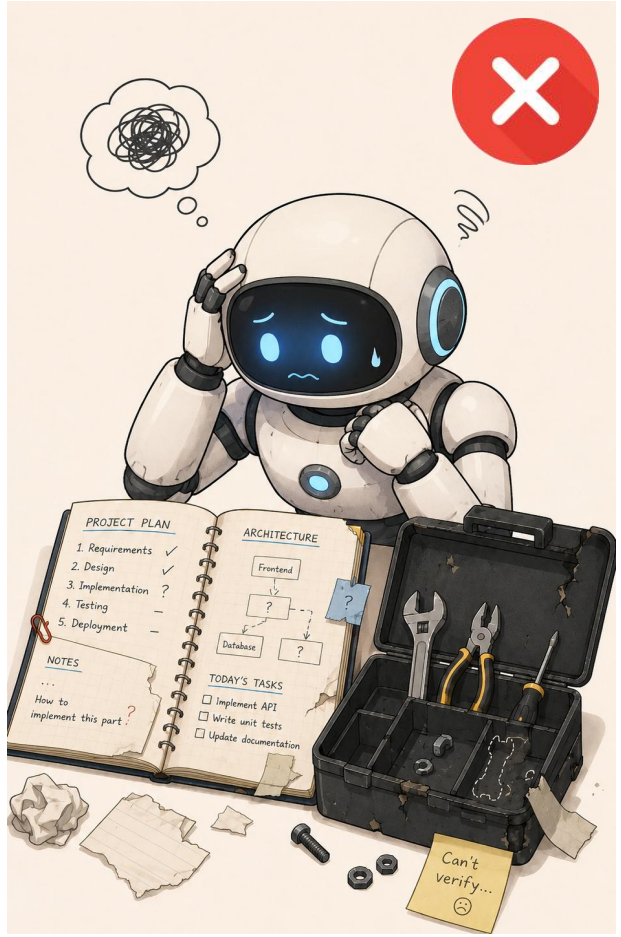
Level 5: Organizations

Organizational AI

OpenAI

Agents Need More than Models

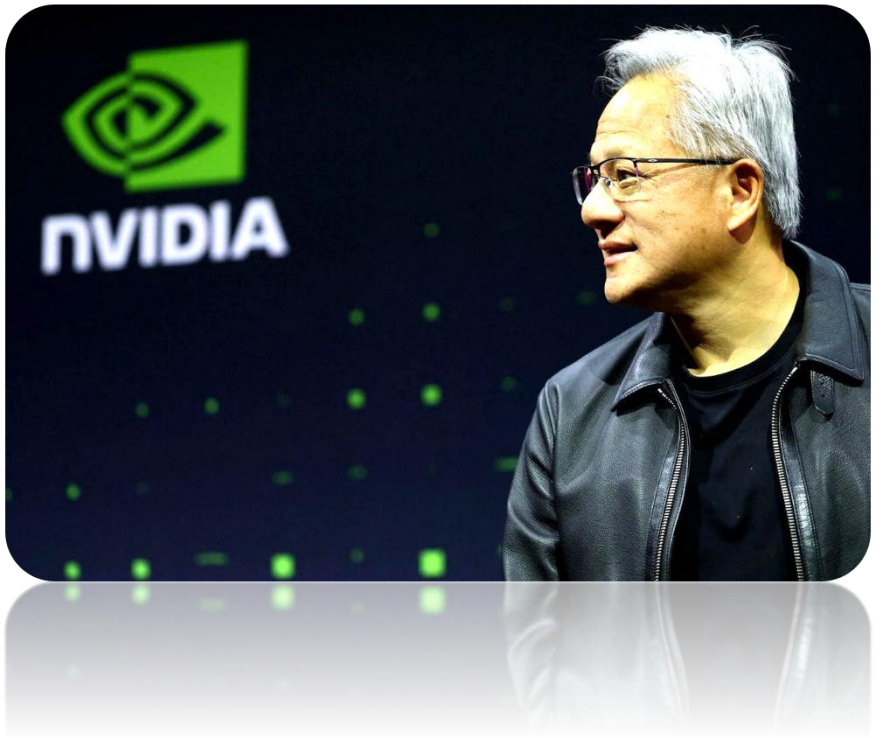




Agent = Model + Harness

> Today, most companies are built on business processes. In the future, most companies will be built on harnesses.

Jensen Huang
Founder and CEO, NVIDIA

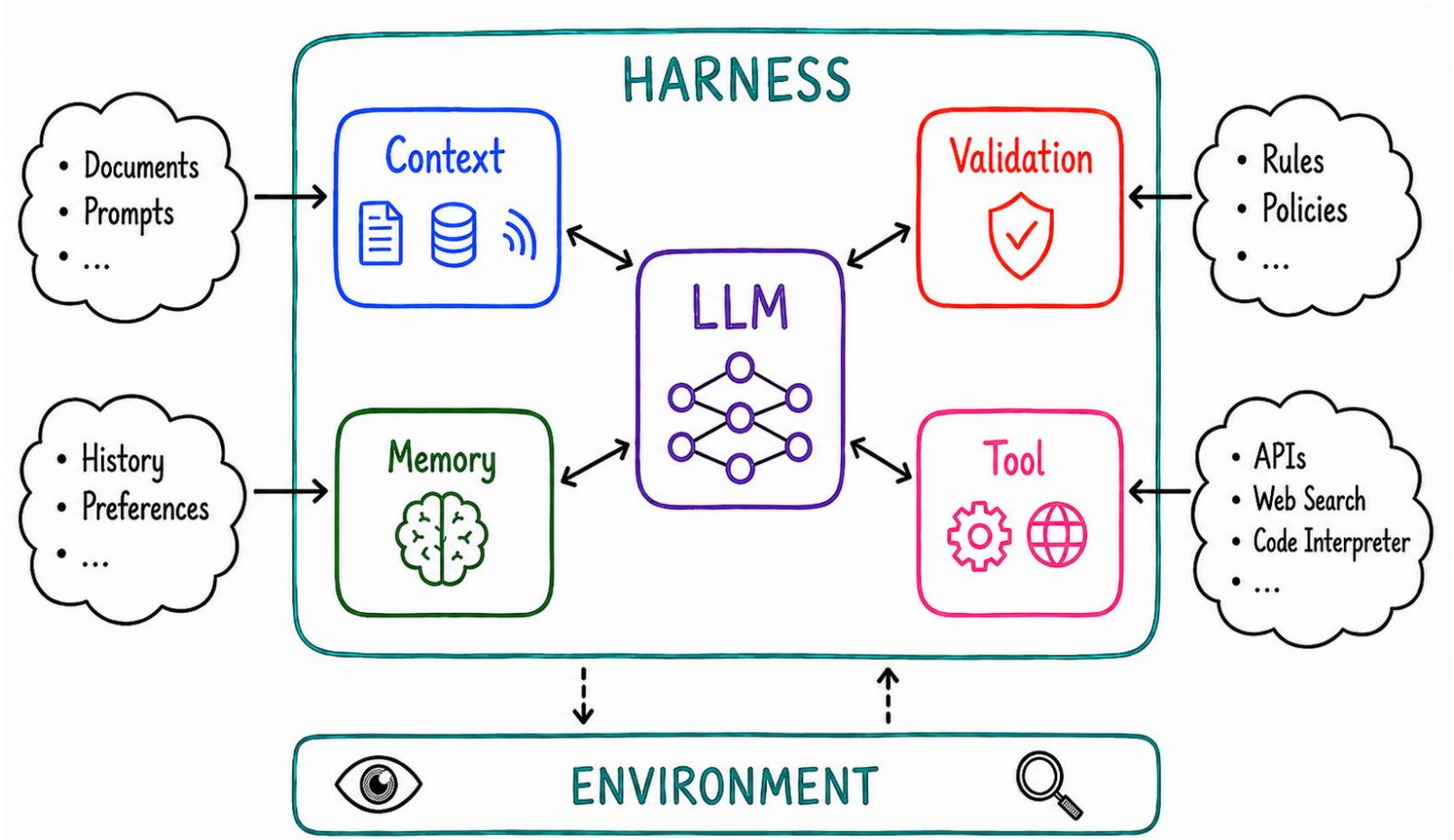


Harness Connects Models and Environments







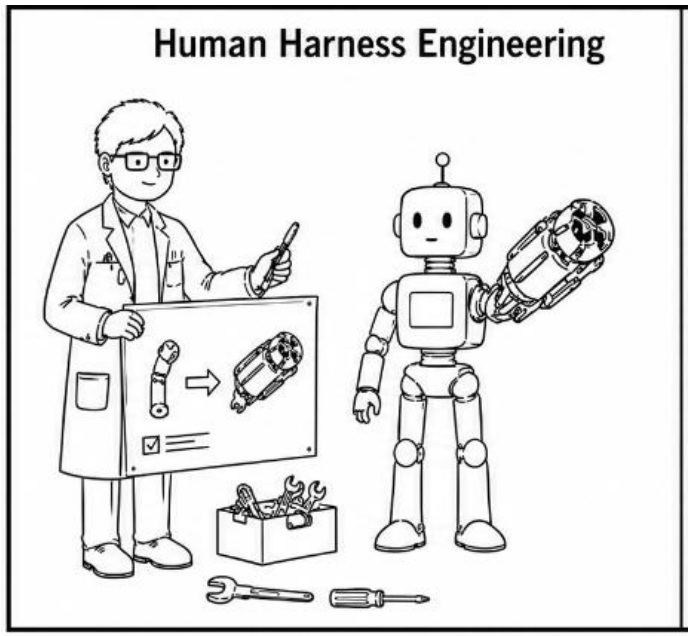


02

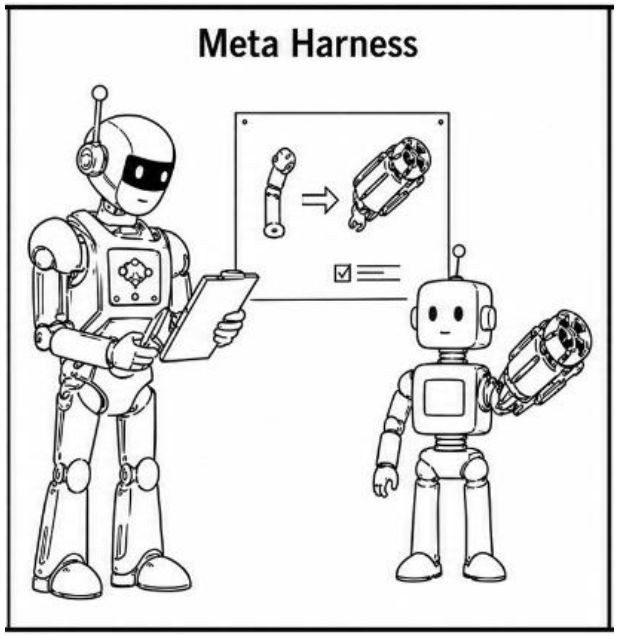
Self-Harness



Harness Engineering



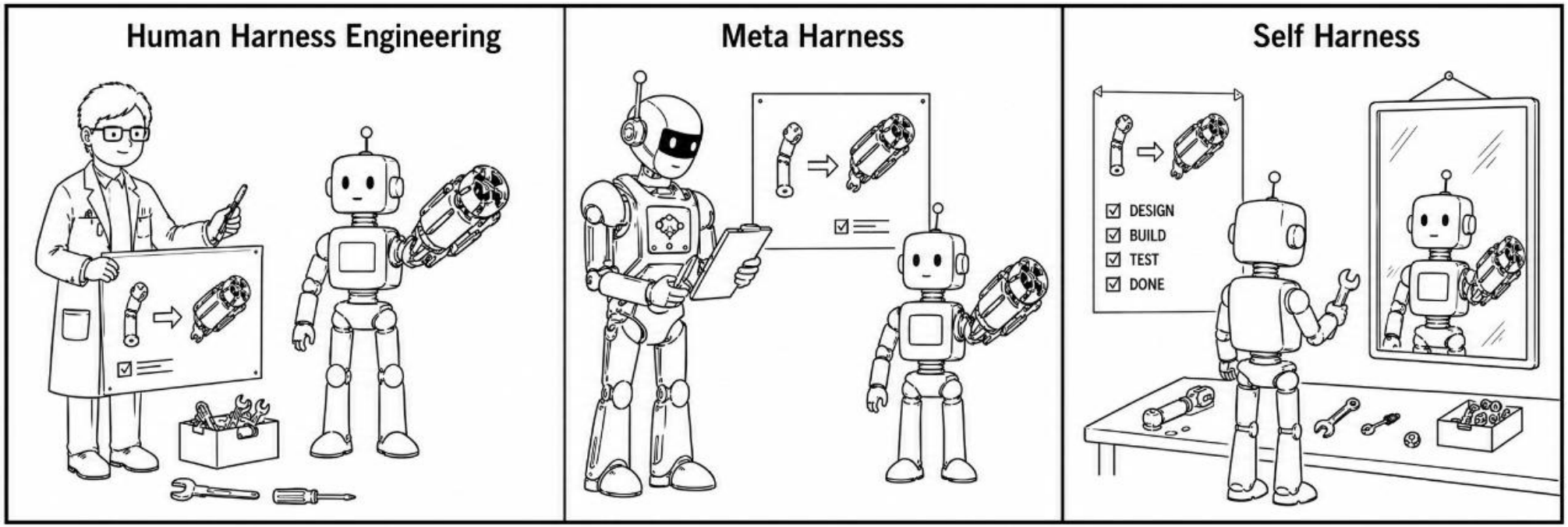
Not fit well with the diversity and rapid evolution of LLMs



Stronger model is costly, unavailable for frontier models, or does not match the target model's failure modes

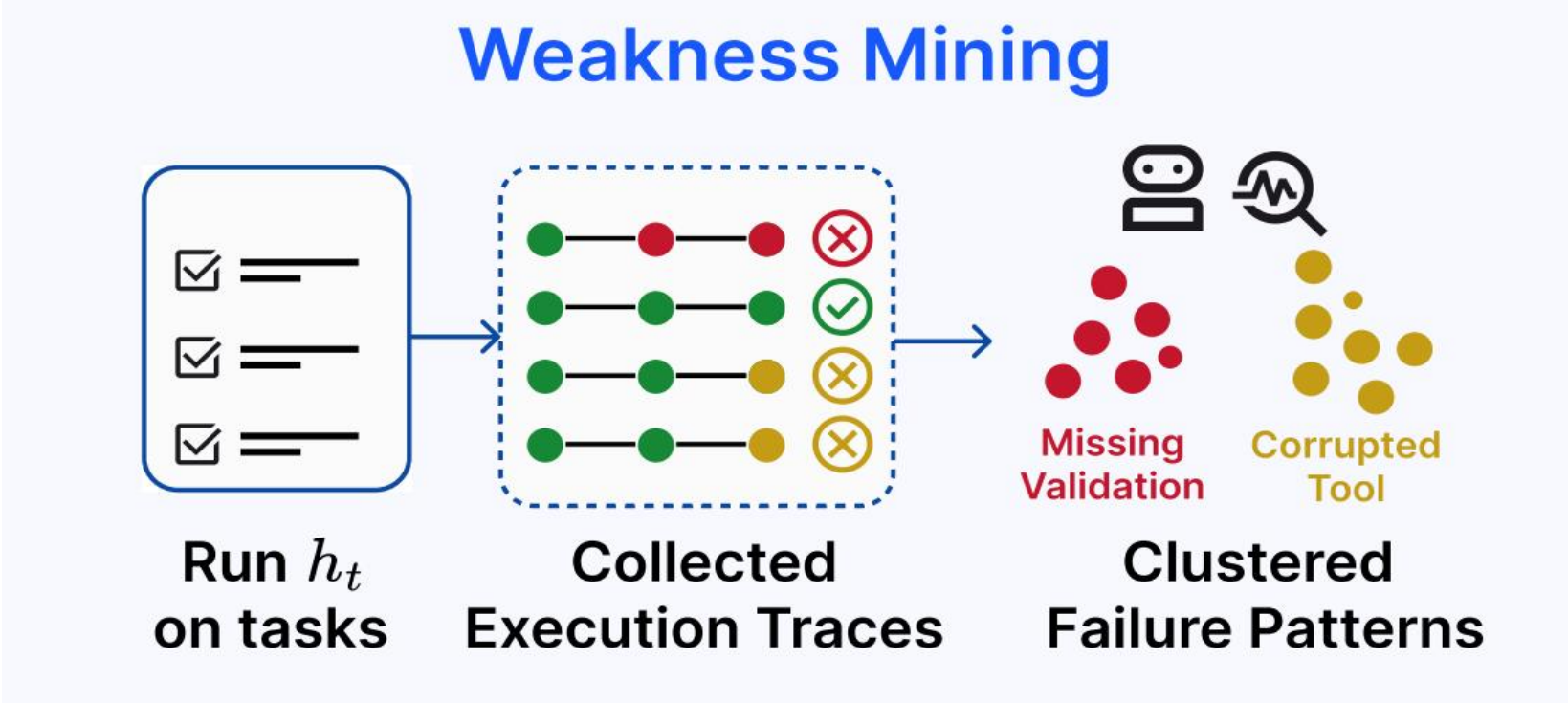
Zhang, H., Zhang, S., Li, K., Zhang, C., Chen, Y., Zhang, Y., Bai, L., & Hu, S. (2026). Self-Harness: Harnesses That Improve Themselves. arXiv preprint arXiv:2606.09498.

Self-Harness: Harnesses That Improve Themselves



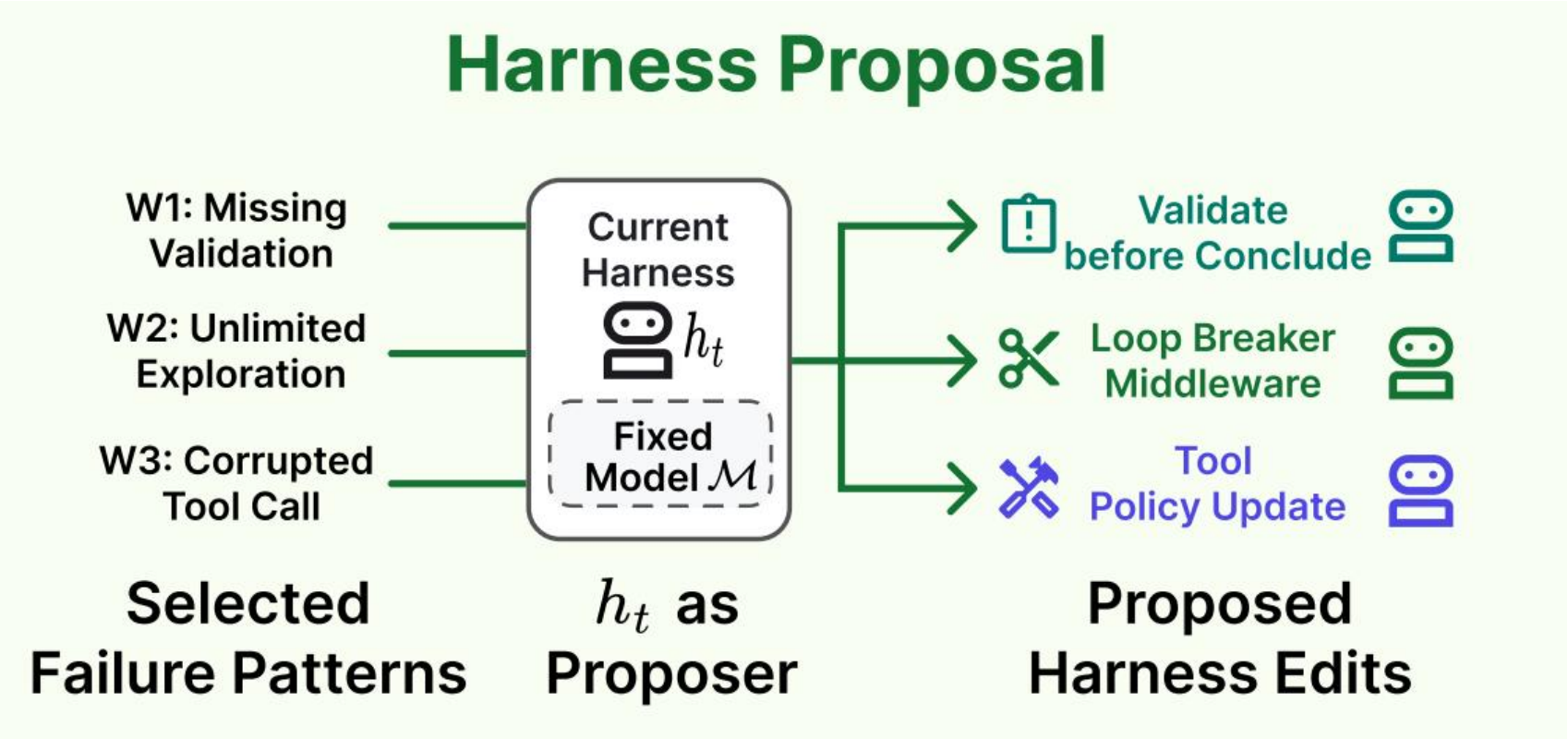
Self-Harness: Harnesses That Improve Themselves

- Cluster failed traces to identify **recurring, model-specific weaknesses** and **failure mechanisms**.



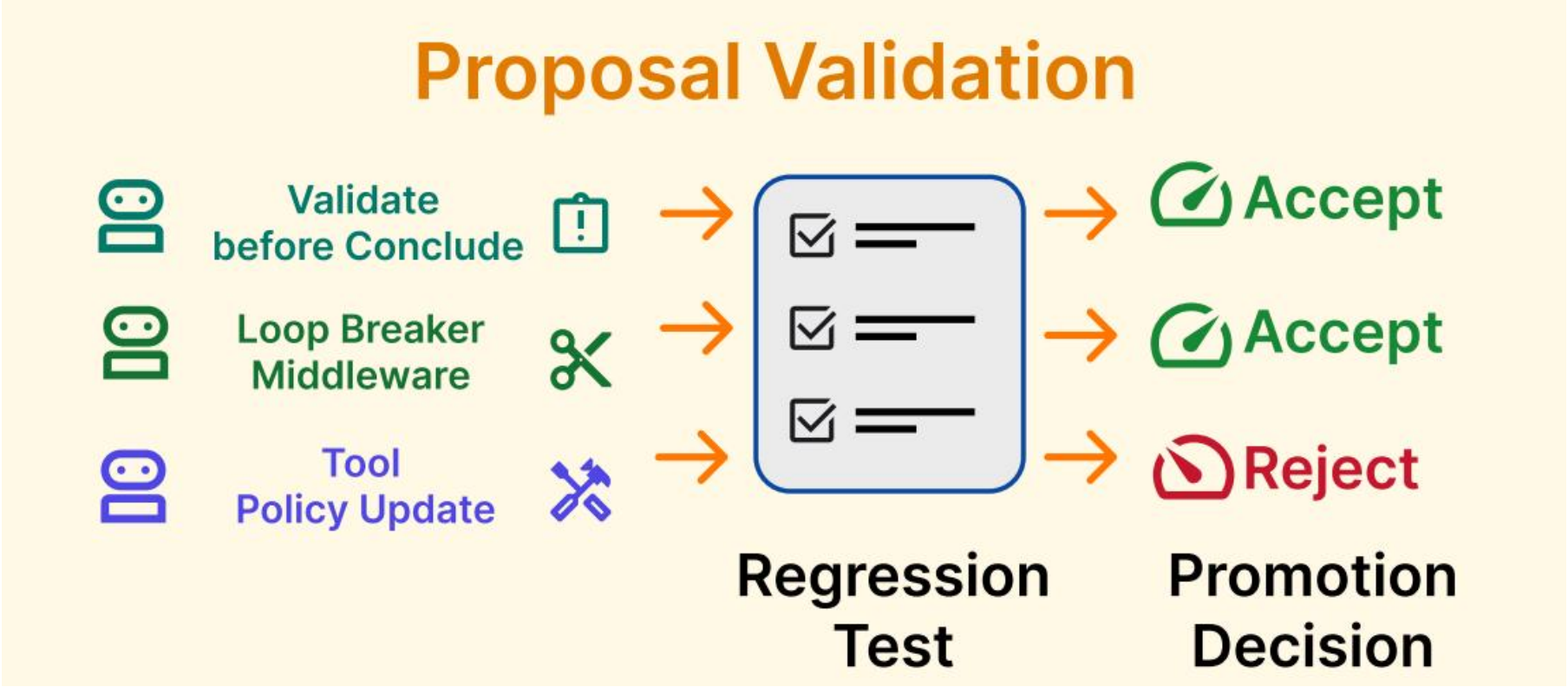
Self-Harness: Harnesses That Improve Themselves

- Turn failures **into diverse, minimal edits** targeting **specific failure mechanisms.**

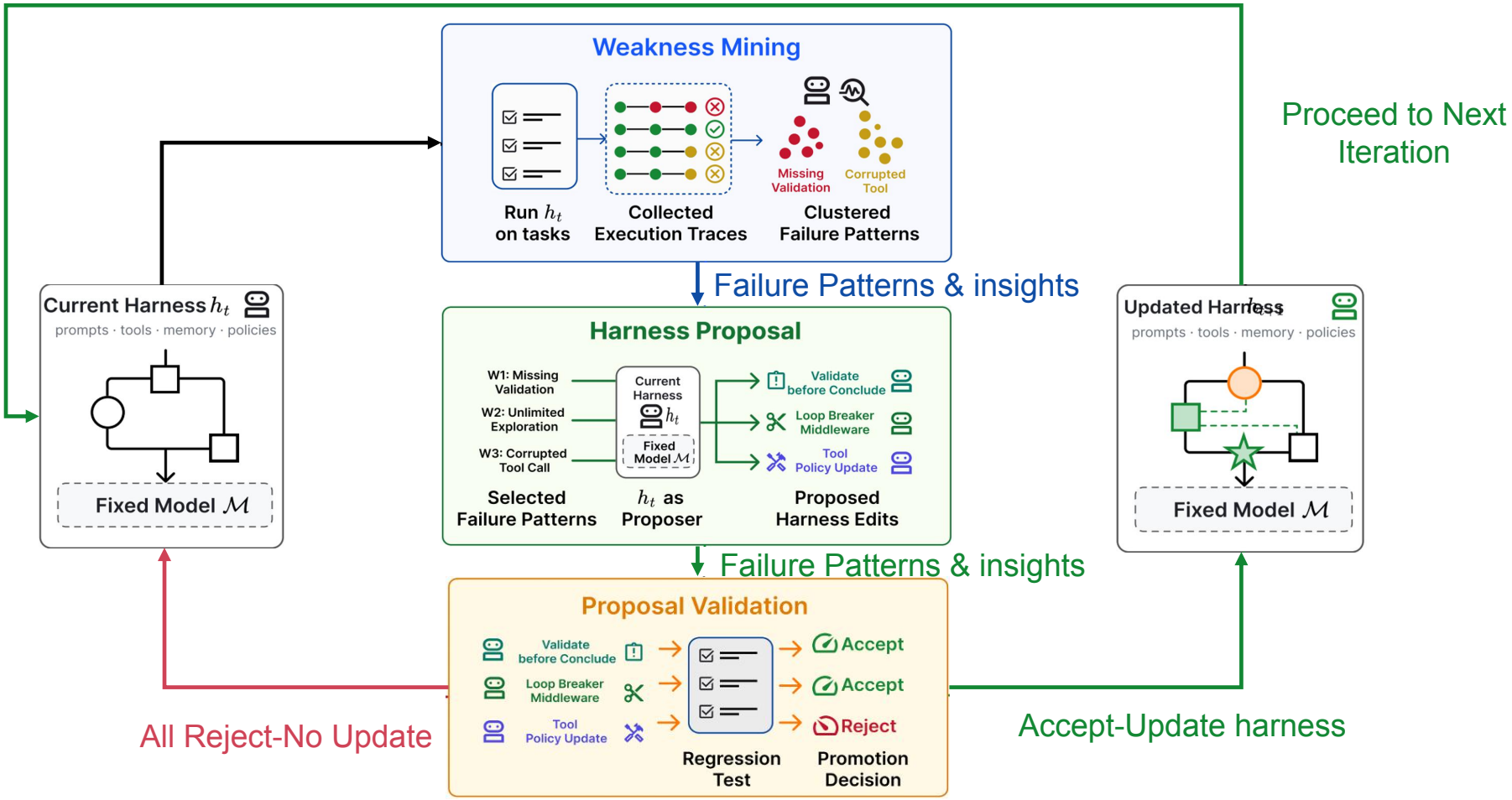


Self-Harness: Harnesses That Improve Themselves

- Regression-test each candidate, **keeping only edits that improve performance**.

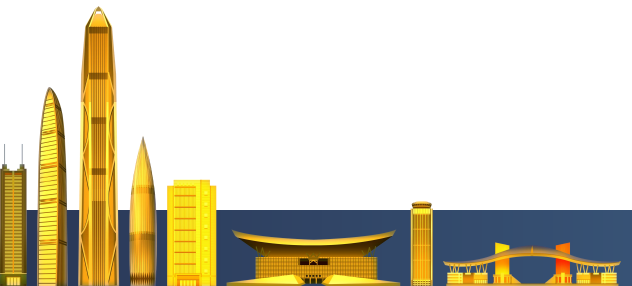


Self-Harness: Harnesses That Improve Themselves

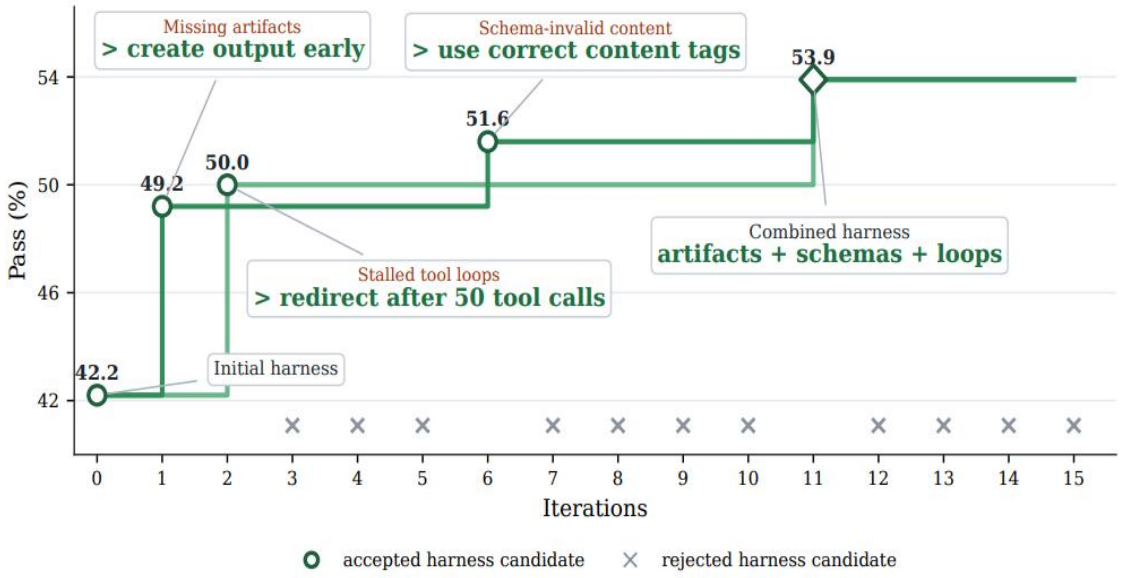


Up to 132% Gains

Model	Initial harness			Self-Harness		
	Held-in	Held-out	Overall	Held-in	Held-out	Overall
Terminal-Bench-2.0						
MiniMax M2.5	43.0	40.5	42.2	**50.0** *(+16%)*	**61.9** ***(+53%)***	**53.9** *(+28%)*
Qwen3.5-35B-A3B	15.1	23.8	18.0	**36.0** ***(+138%)***	**38.1** ***(+60%)***	**36.7** ***(+104%)***
GLM-5	47.7	42.9	46.1	**57.0** *(+20%)*	**57.1** *(+33%)*	**57.0** *(+24%)*
SWE-bench Verified						
MiniMax M2.5	51.5	34.8	46.0	**58.2** *(+13%)*	**40.9** *(+18%)*	**52.5** *(+14%)*
Qwen3.5-35B-A3B	20.1	18.2	19.5	**42.5** ***(+111%)***	**39.4** ***(+116%)***	**41.5** ***(+113%)***
GLM-5	53.7	48.5	52.0	**58.2** *(+8%)*	**50.0** *(+3%)*	**55.5** *(+7%)*
AppWorld						
MiniMax M2.5	51.7	45.6	48.6	**62.8** *(+21%)*	**55.0** *(+21%)*	**58.9** *(+21%)*
Qwen3.5-35B-A3B	25.0	20.0	22.5	**60.0** ***(+140%)***	**44.4** ***(+122%)***	**52.2** ***(+132%)***
GLM-5	47.8	41.1	44.4	**92.2** ***(+93%)***	**77.8** ***(+89%)***	**85.0** ***(+91%)***

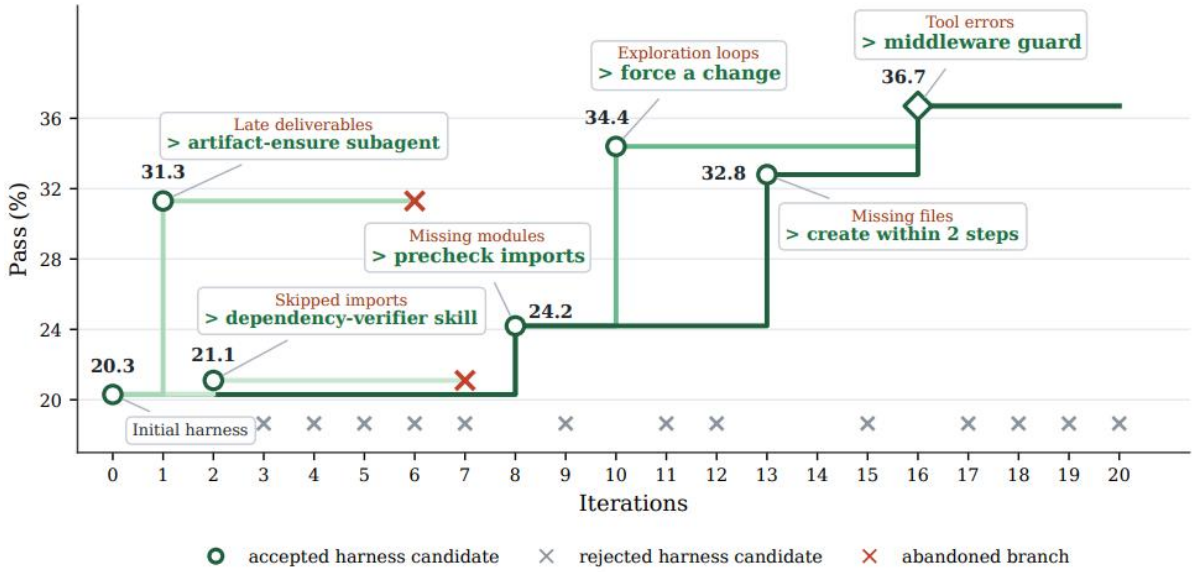


Harness Evolution Targeting Model-Specific Weaknesses



Minimax M2.5

Limited exploration + fix file tags



Qwen 3.5 35B-A3B

Block duplicate calls + check dependencies + recover outputs

Community Feedback



Lilian Weng

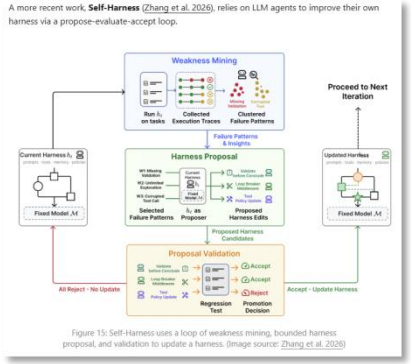
OpenAI

副总裁，人工智能安全与机器人技术，应用研究 @OpenAI

Harness Engineering for Self-Improvement

The concept of recursive self-improvement (RSI) dates back to I. J. Good (1965), where he defined an "ultraintelligent machine" as a system that can surpass humans in all intellectual activities and design bette...

Date: July 4, 2026 | Estimated Reading Time: 31 min | Author: Lilian Weng





Harrison Chase

Co-Founder and CEO at LangChain

美国 加利福尼亚 旧金山 • 联系方式

53,407 位关注者 • 500+ 好友

LangChain

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Harrison Chase @hwchase17

🧠 Self-Harness: Harnesses that improve themselves

New paper on agents shaping their own harnesses to improve over time. Not from LangChain, but builds on top of DeepAgents! Three key steps:

1/ Weakness mining: find failure modes from traces
2/ Harness proposal: suggest changes to the harness
3/ Proposal validation: does regression testing on proposals and then accepts

Paper: arxiv.org/pdf/2606.09498

Builds on top of DeepAgents: github.com/langchain-ai/d...

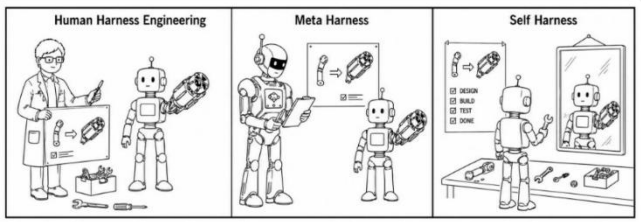


Figure 1: Three paradigms of harness improvement. In human harness engineering, human engineers manually revise the agent harness. In Meta-Harness, a stronger external agent guides the improvement of a weaker target agent. In Self-Harness, the agent improves its own operating harness.

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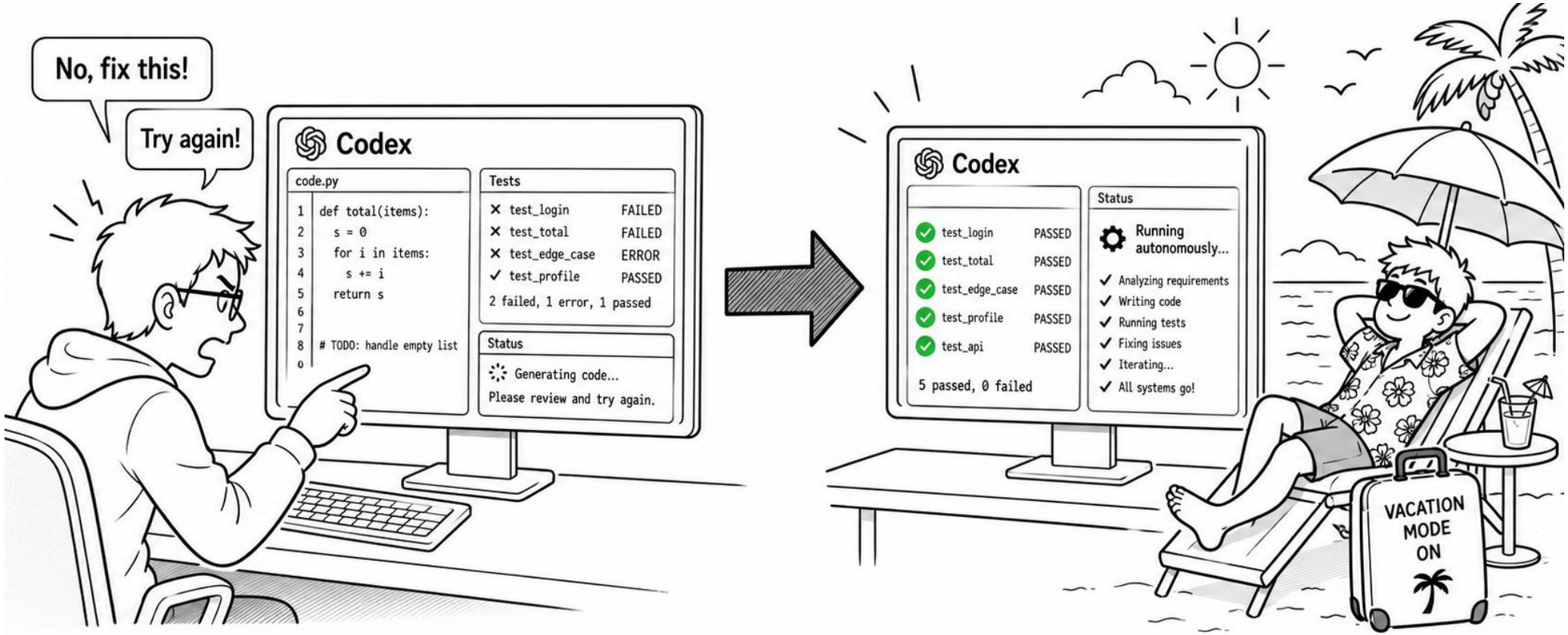


03

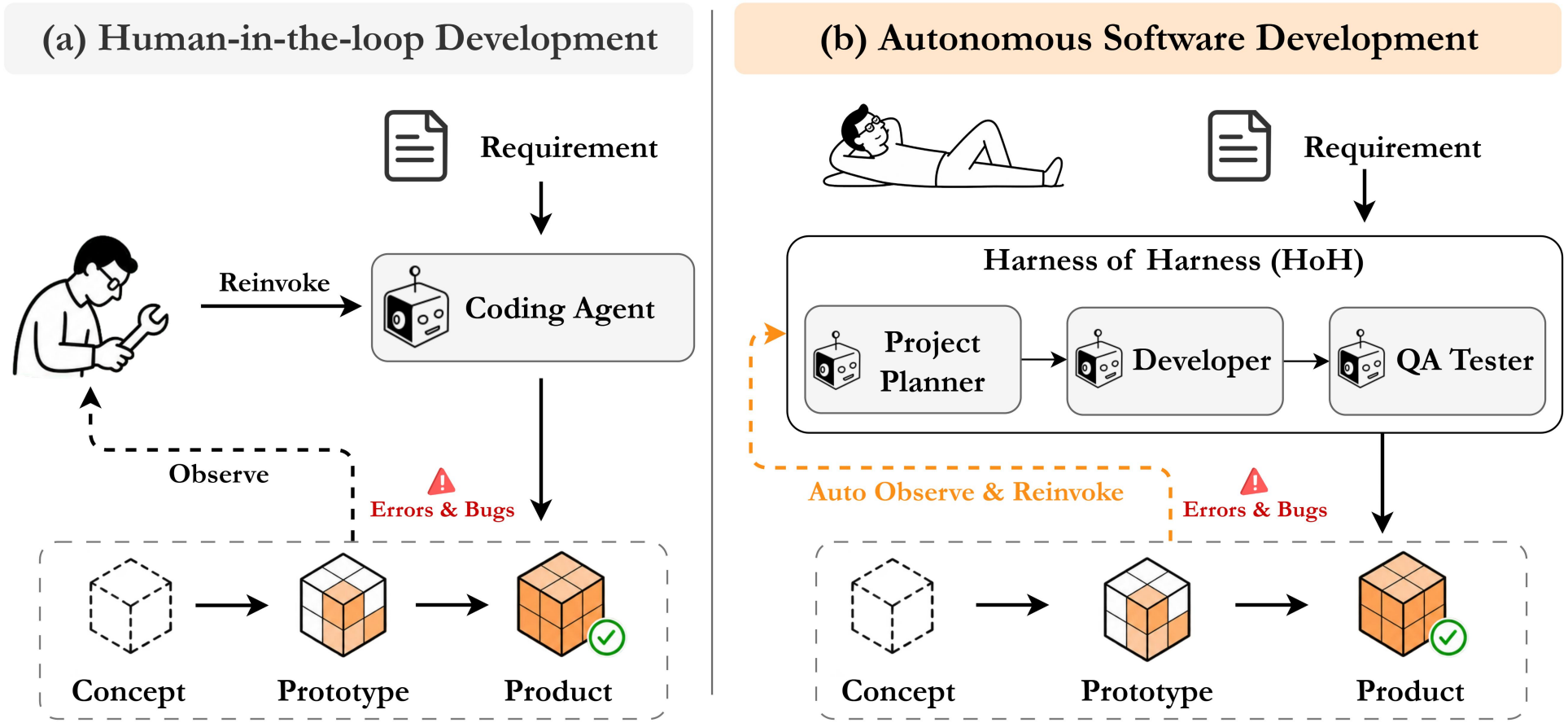
Harness of Harness



Human Supervision and Autonomous Development



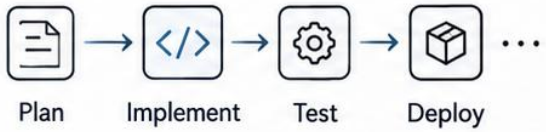
Two Development Modes



Why Autonomous Development Is Hard

1 Long-Horizon Coordination

1.1 Multi-stage development

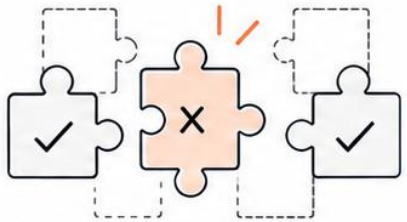


1.2 Lose earlier context

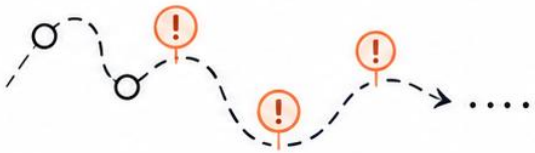


2 Global Consistency

2.1 Local fixes break global constraints

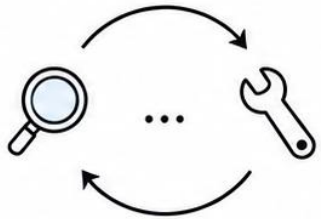


2.2 Errors accumulate over time



3 Reliable Self-Verification

3.1 Repeated inspection and repair



3.2 Premature task termination

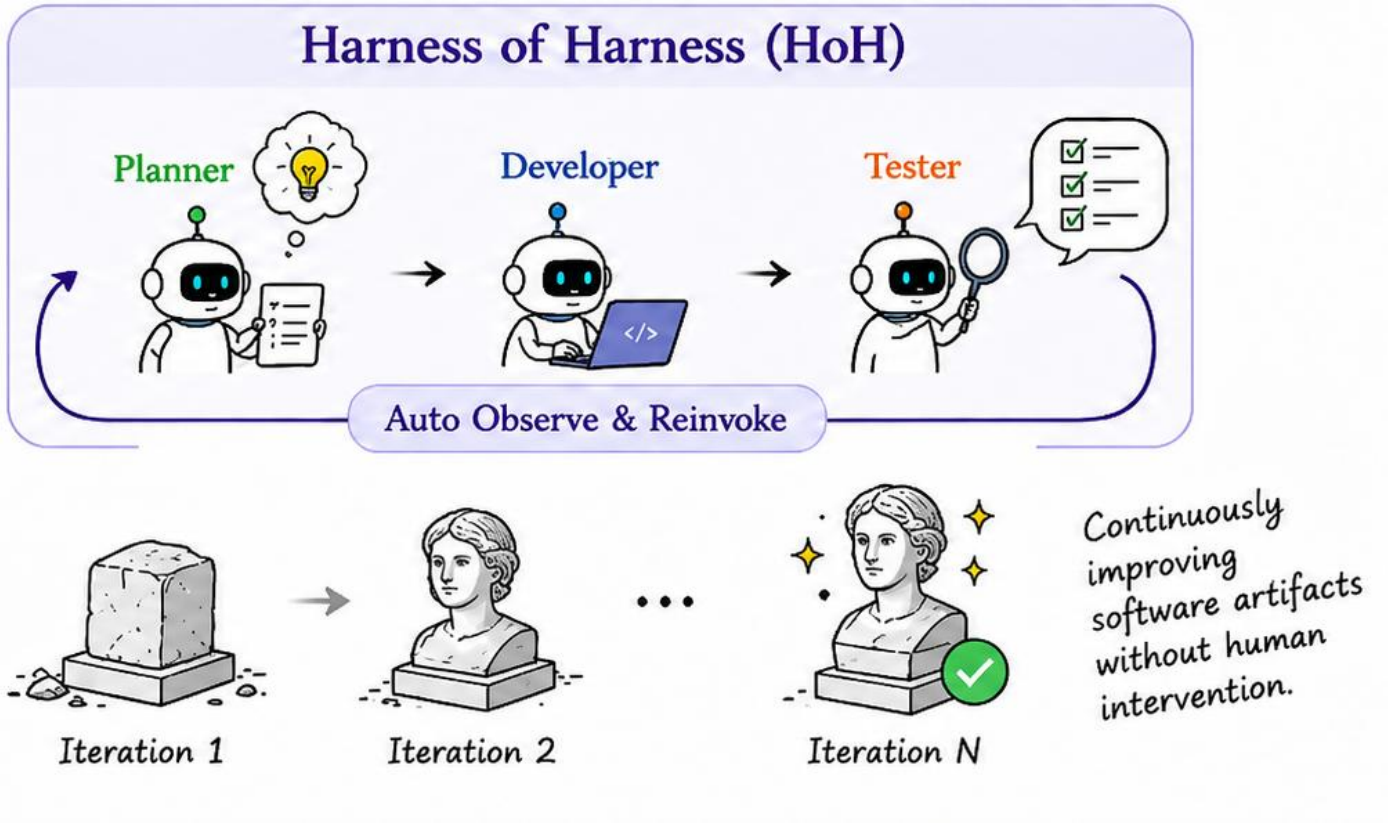


Harness of Harness: Continual Improvement

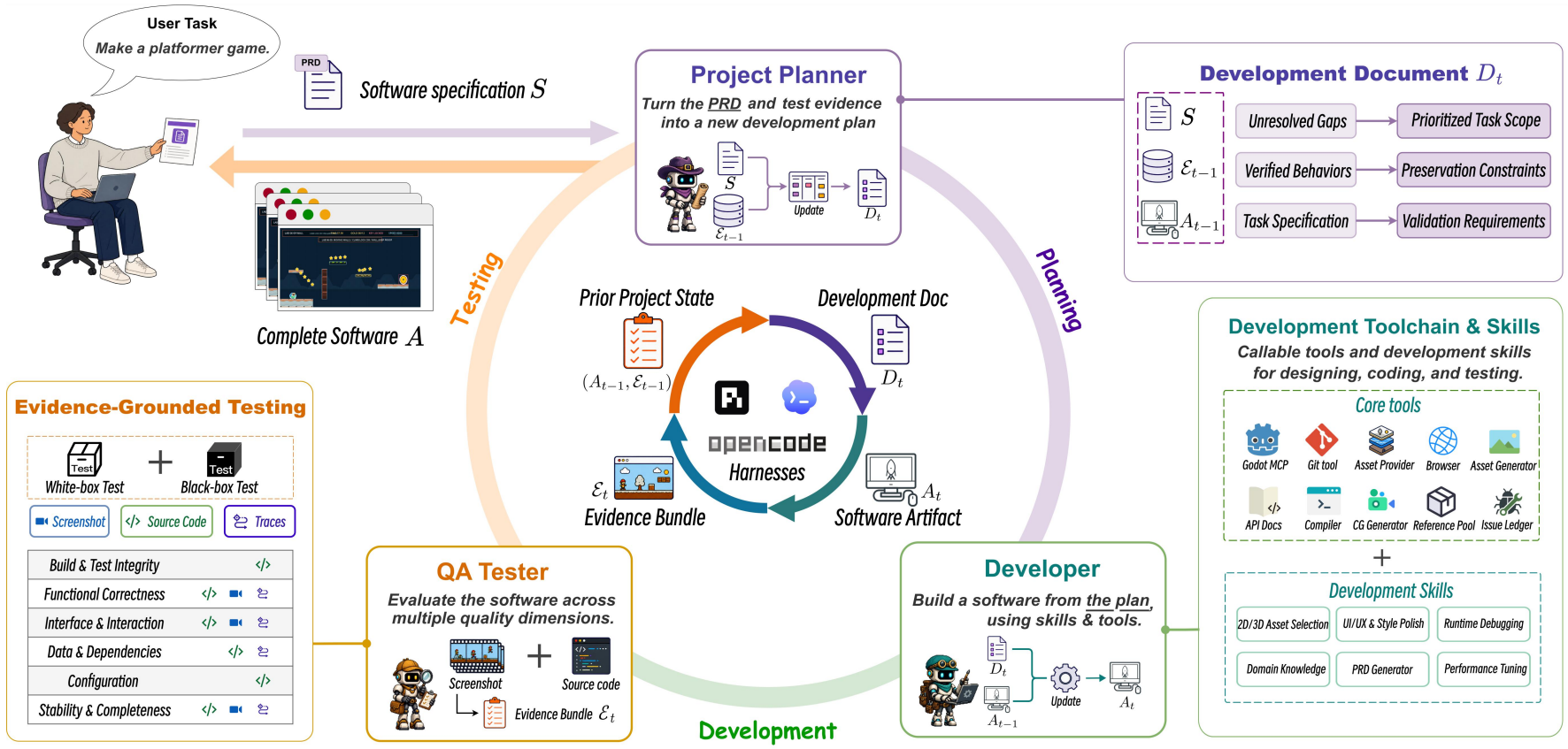


Focus on bigger things!





Harness of Harness: Enabling Continual Improvement

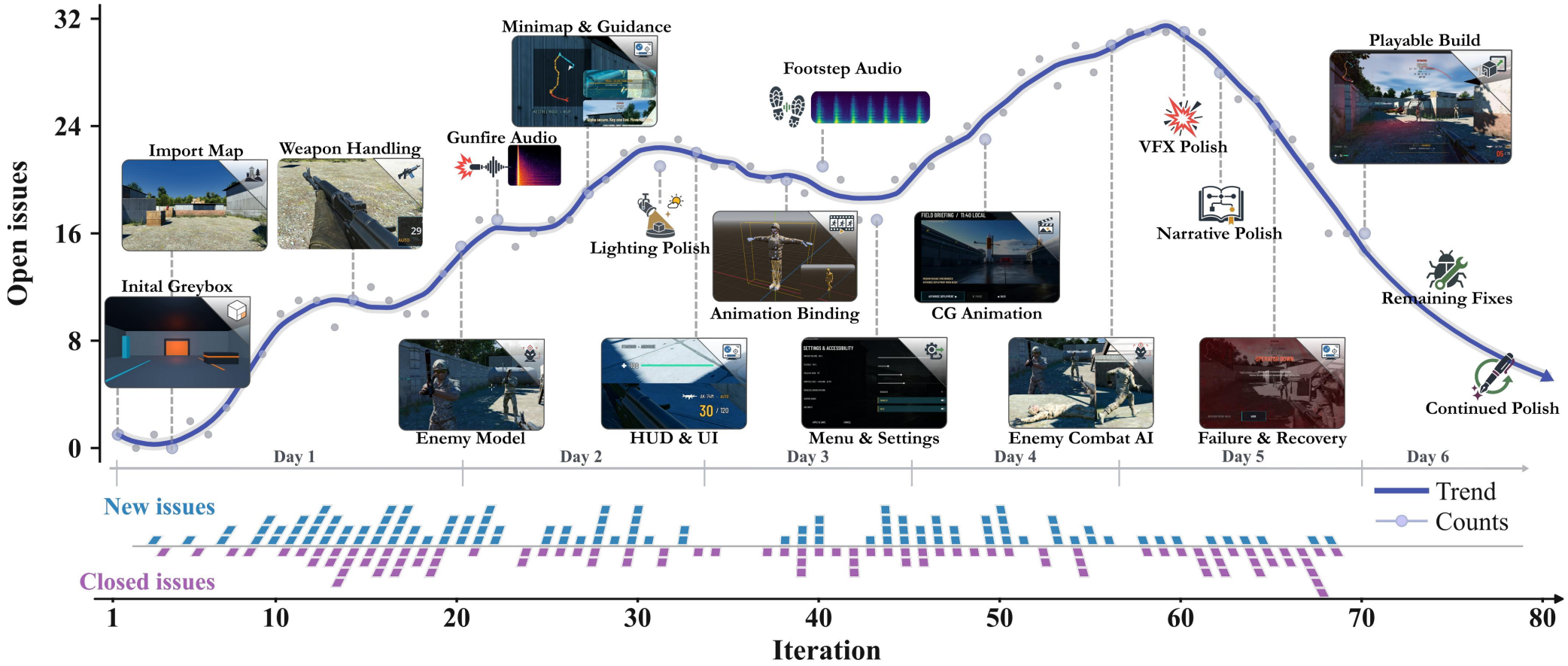


Consistent Gains across Benchmarks and Harnesses

Setting	GameCraft-Bench						FrontierSWE				ProgramBench
	Action	Timing	Strat.	Sim.	Adv.	***Overall***	Impl.	Perf.	Research	***Dominance***	***Pass Rate***†
						Codex + GPT-5.5 (high)					
Vanilla	48.74	48.80	44.06	53.63	52.68	49.58	0.21	0.17	1.15	44%	60.41
HoH@1	59.73	53.79	57.01	66.75	61.24	59.71	0.24	0.44	1.30	58%	65.42
HoH@2	64.34	62.03	59.97	71.77	66.11	64.84	0.28	0.45	1.18	60%	65.79
HoH@3	**71.02**	**70.26**	**66.13**	**78.42**	**71.76**	**71.52** (+21.93)	**0.30**	**0.45**	**1.45**	**71%** (+27)	**66.50** (+6.09)
						OpenCode + DeepSeek-V4-Pro					
Vanilla	26.21	24.05	21.27	37.12	25.84	26.90	0.08	0.23	0.57	25%	45.27
HoH@1	27.75	27.40	21.73	43.75	22.44	28.61	0.09	0.23	0.78	28%	55.33
HoH@2	43.22	36.56	33.51	52.26	36.05	40.32	0.10	0.27	0.78	42%	55.66
HoH@3	**49.00**	**45.05**	**43.34**	**55.86**	**51.64**	**48.98** (+22.08)	**0.15**	**0.27**	**0.78**	**44%** (+19)	**57.56** (+12.29)
						Pi + MiniMax-M3					
Vanilla	45.64	38.20	34.53	48.19	44.26	42.16	0.06	0.12	1.30	35%	35.83
HoH@1	50.59	52.23	38.60	54.00	49.89	49.06	0.10	0.42	**1.89**	62%	48.68
HoH@2	54.70	56.52	42.33	63.20	58.47	55.04	0.11	0.43	**1.89**	**66%**	**53.57**
HoH@3	**58.24**	**62.10**	**44.86**	**64.25**	**64.44**	**58.78** (+16.62)	**0.11**	**0.45**	1.88	64% (+29)	52.68 (+16.85)



Multi-Day Development of a FPS Game

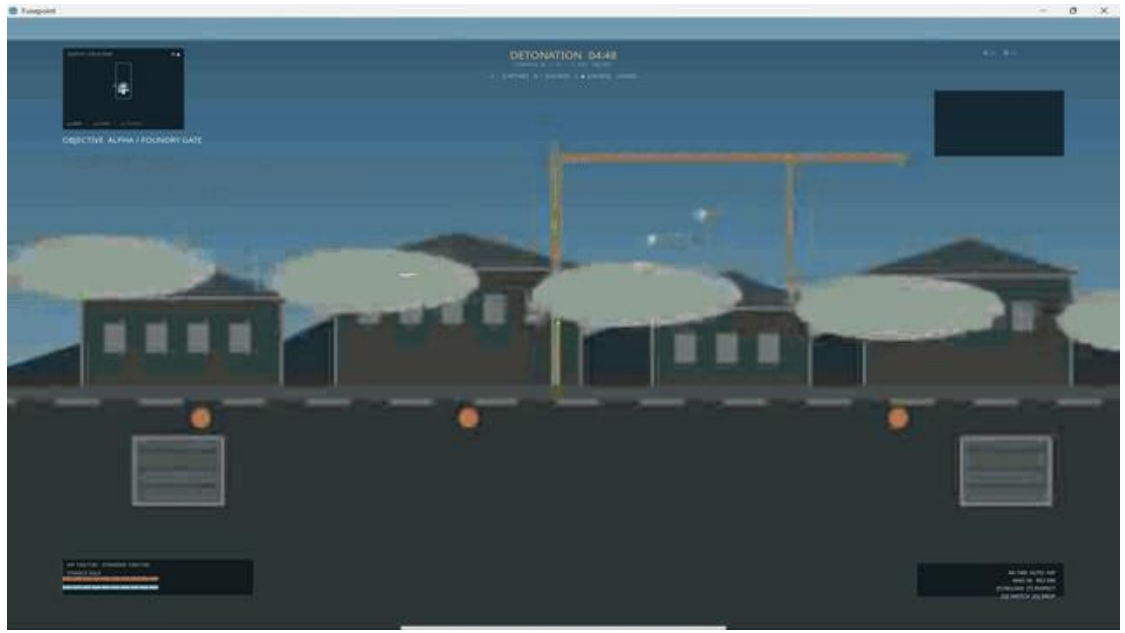


Harness of Harness vs. /goal

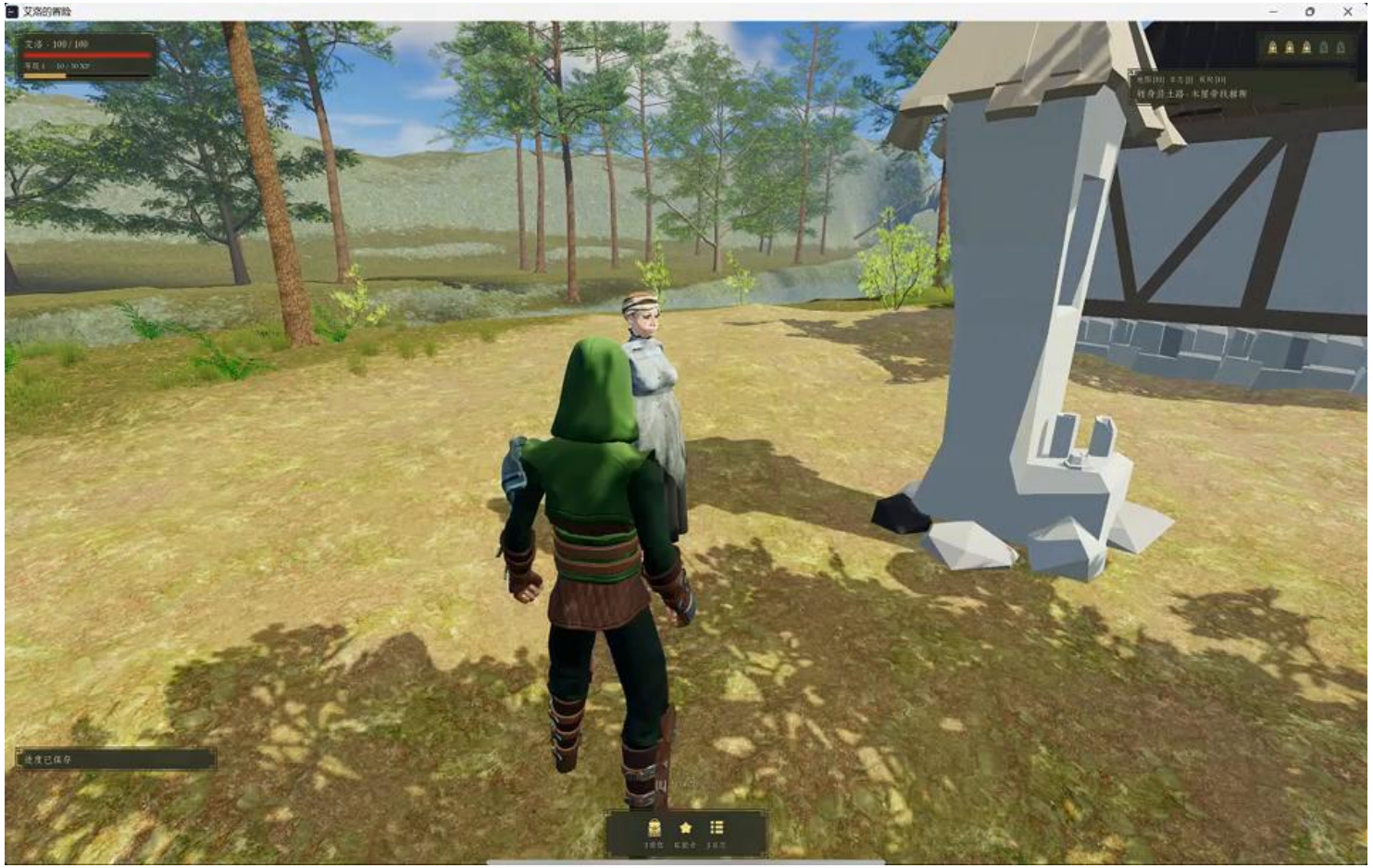
Same PRD and Codex (GPT 5.6)



Harness of Harness



/goal


艾洛的冒险
艾洛 - 100 / 100
进度已保存

Harness of Harness:

Multi-Day Autonomous Software Development with Continual Improvement

Haoyang Yan · Minle Su · Hangfan Zhang · Zhanhao Li · Chen Zhang · Shao Zhang · Yang Chen · Lei Bai · Shuyue Hu
Shanghai Artificial Intelligence Laboratory

Flesymeb/GameLoop

flesymeb.github.io/GameLoop

arxiv.org/abs/XXXX.XXXXX



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Thanks!



Harness of Harness: Enabling Continual Improvement

balances repair with capability growth

scopes development into small and verifiable increments

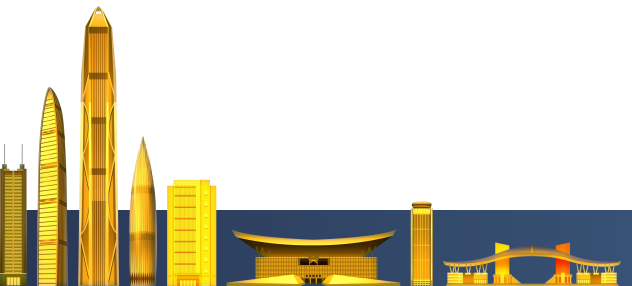
separates implementation-time testing from independent evaluation

constrains verifiable outputs rather than prescribing agent workflows

progressively exposes deliverables, role-specific tools, and skills

encourages reuse rather than recreation

maintains versioned project histories



Harness of Harness vs. /goal

Same PRD and Codex (GPT 5.6)



Harness of Harness



/goal